Building a carbon rotor blade

Sandwich construction with vacuum infusion

ROTOR BLADE MANUFACTURING – Erasmus – 18.11.2026

Name:		**Class:**	Erasmus
Date:		**Grade:**	

What we build

A half-pipe shell made of **two carbon layers with a foam core in between**. Several shells will become the rotor blades of a wind turbine built on one bicycle rim. The shape comes from a **pipe** that serves as the mould. The resin is not brushed on: it is **drawn into the dry lay-up by vacuum** (vacuum infusion).

> △ **Safety**
> Wear **nitrile gloves, safety glasses and an apron** when you handle resin and hardener. Make sure the room is well ventilated. Carbon fibres itch: do not rub your eyes.

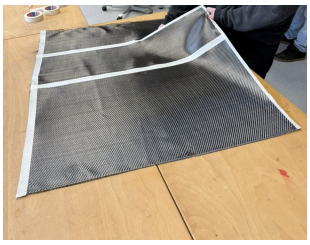
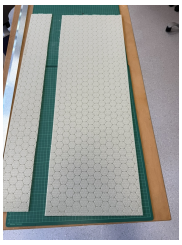
Layer stack

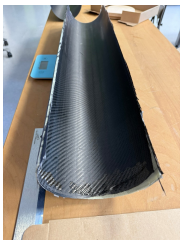
#	Layer	Size	Function
1	Carbon	85 cm × 30 cm	Structural layer, visible side on the outside
2	Core: 3DCore PET 100, 3 mm	85 cm × 30 cm	Makes the blade stiff, lets the resin flow
3	Carbon	85 cm × 30 cm	Second structural layer
4	Peel ply + flow mesh	87 cm × 32 cm	Spreads the resin; peeled off after curing
5	Vacuum bag (tube film)	width 86 cm	Encloses everything, pressed on by air pressure

Steps

1. Tape the carbon along the cutting lines, then cut two pieces. Cut the core and the peel ply + flow mesh.

2. Lay carbon, core (fixed with spray adhesive) and the second carbon layer on the pipe. Put peel ply + flow mesh on top.
3. Slide the lay-up into the vacuum bag. Attach the resin hose and the vacuum hose and seal the connections.
4. Connect the pump and **pull the vacuum**.
5. Mix the resin in **3 portions**: 200 g resin + 60 g hardener (fast) each, ratio $100 : 30$.
6. Open the resin hose: the vacuum draws the resin into the lay-up. Watch the flow front.
7. Cure with the infrared lamp. The blades are ready in the evening or the next day.



Tasks

1. Resin. The ratio resin : hardener is $100 : 30$. How much hardener do you need for 600 g of resin? Why do we mix three small portions instead of one big batch?

2. Vacuum. The lay-up covers about 0,85 m $\times$ 0,30 m. Estimate the force of the air pressure on it if the pump removes about 90 % of the air (air pressure $\approx$ 1000 hPa).

3. Layers. Match each layer to its job: *peel ply* *core* *carbon* *flow mesh*

(a) carries the loads: _____ (b) spreads the resin: _____

(c) gives stiffness at low weight: _____ (d) leaves a rough surface for gluing: _____

4. The trial blade. Our trial blade was built with 600 g resin and 180 g hardener and weighs 634 g. How many grams of resin did you use for your blade (3 portions)? Is the finished blade lighter or heavier than the resin you mixed? Give one reason.

	Trial blade (workshop recipe)
Carbon	2 layers
Core	3DCore PET 100, 3 mm
Resin + hardener	600 g + 180 g
Finished blade	634 g

5. Your blade. Weigh your finished blade and measure it.

Quantity	**Your value**
Mass	g
Length / width	cm × cm
Comparison with trial B	